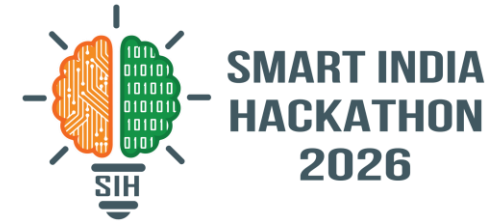
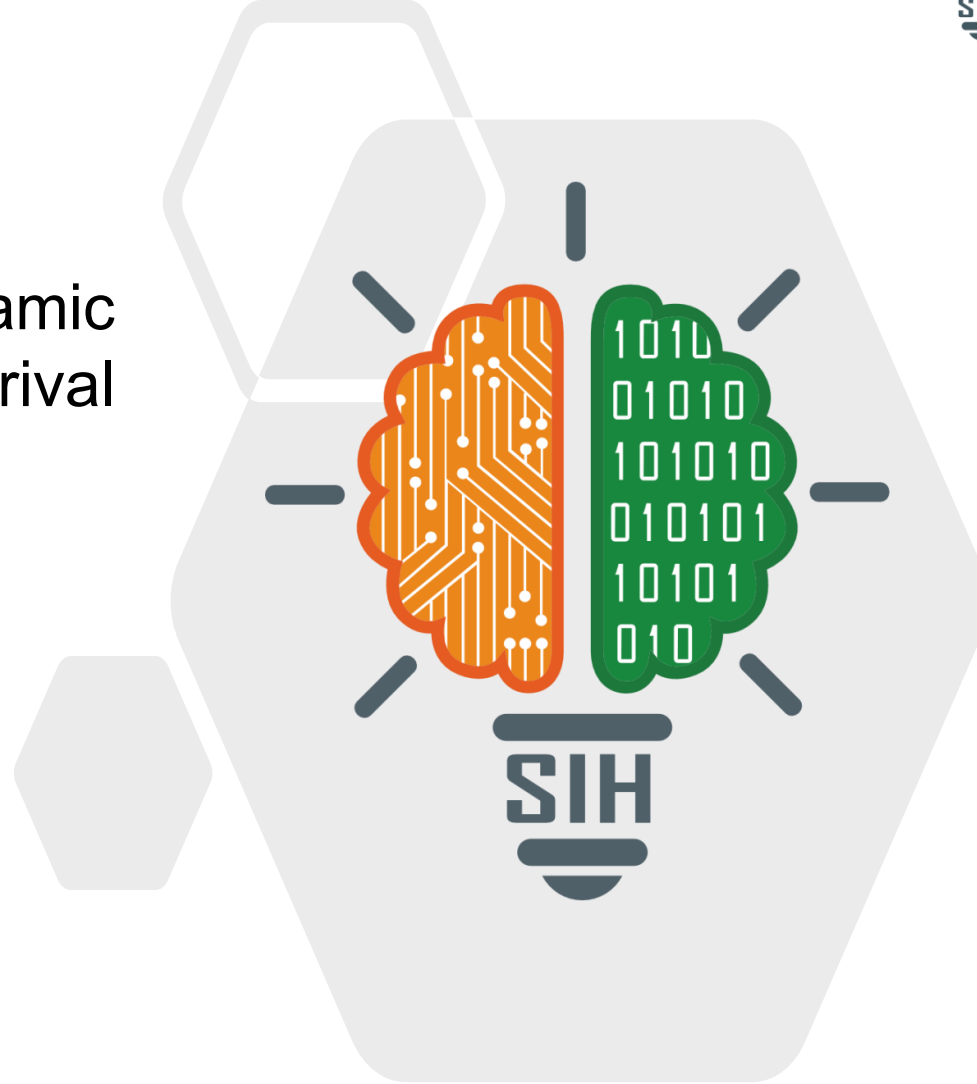


SMART INDIA HACKATHON 2026



- **Problem Statement ID –** SIH26028
- **Problem Statement Title-** Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name -** TypeError-4-ever



❖ Proposed Solution forecast the cause, not the clock

PROBLEM AT HAND

Delay is repeated, not forecast

Today's ETA carries the current delay forward and trusts timetable padding. It cannot see congestion building ahead.

The network goes blind between stations

Position is reported only at instrumented stations - gaps of 40–60 minutes on long sections.

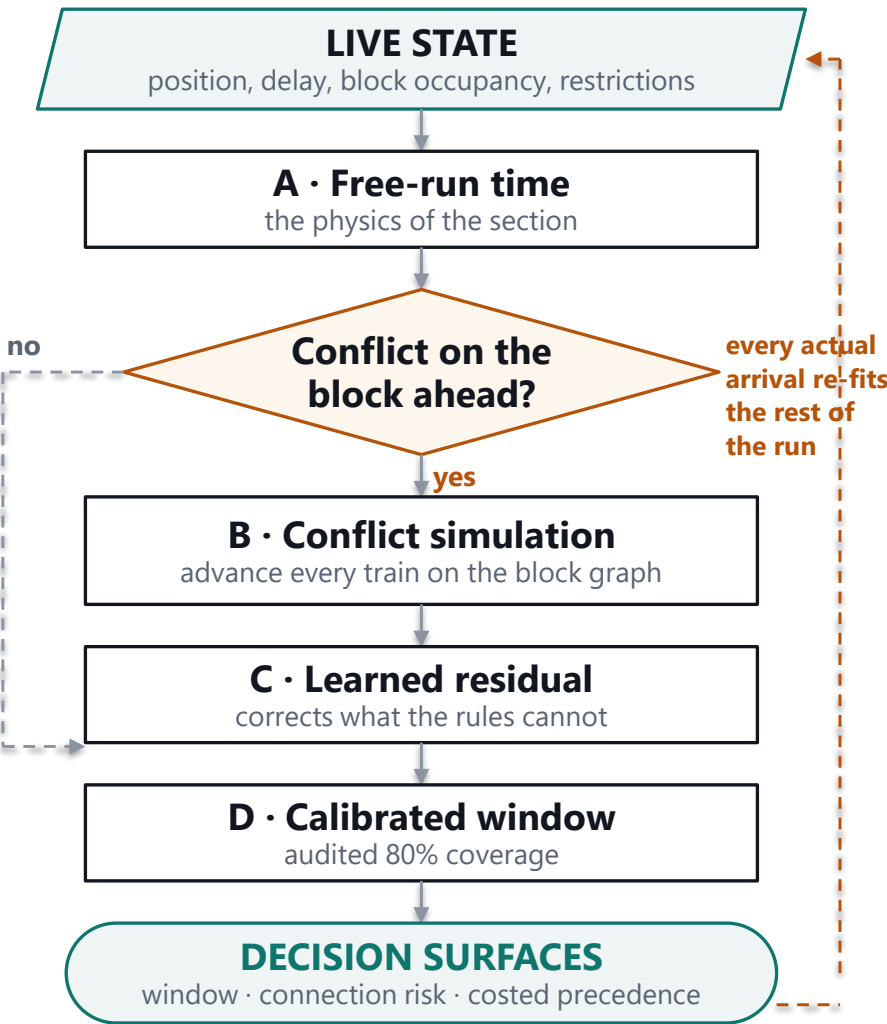
Cascades are invisible

One train held at a junction delays every train behind it, yet no deployed ETA model represents that dependency.

One number hides the risk

No confidence attached, so nobody knows when it is safe to act - and nothing handles a diversion mid-run.

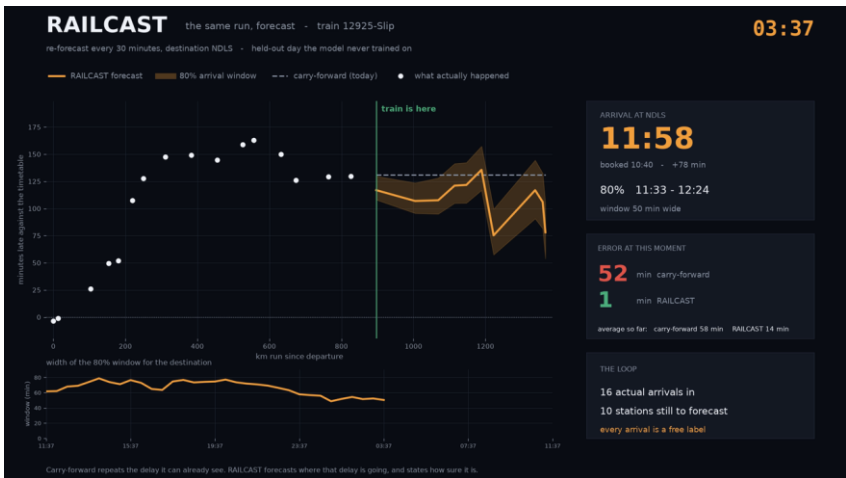
HOW ONE FORECAST IS MADE



PROTOTYPE (RUNNING)

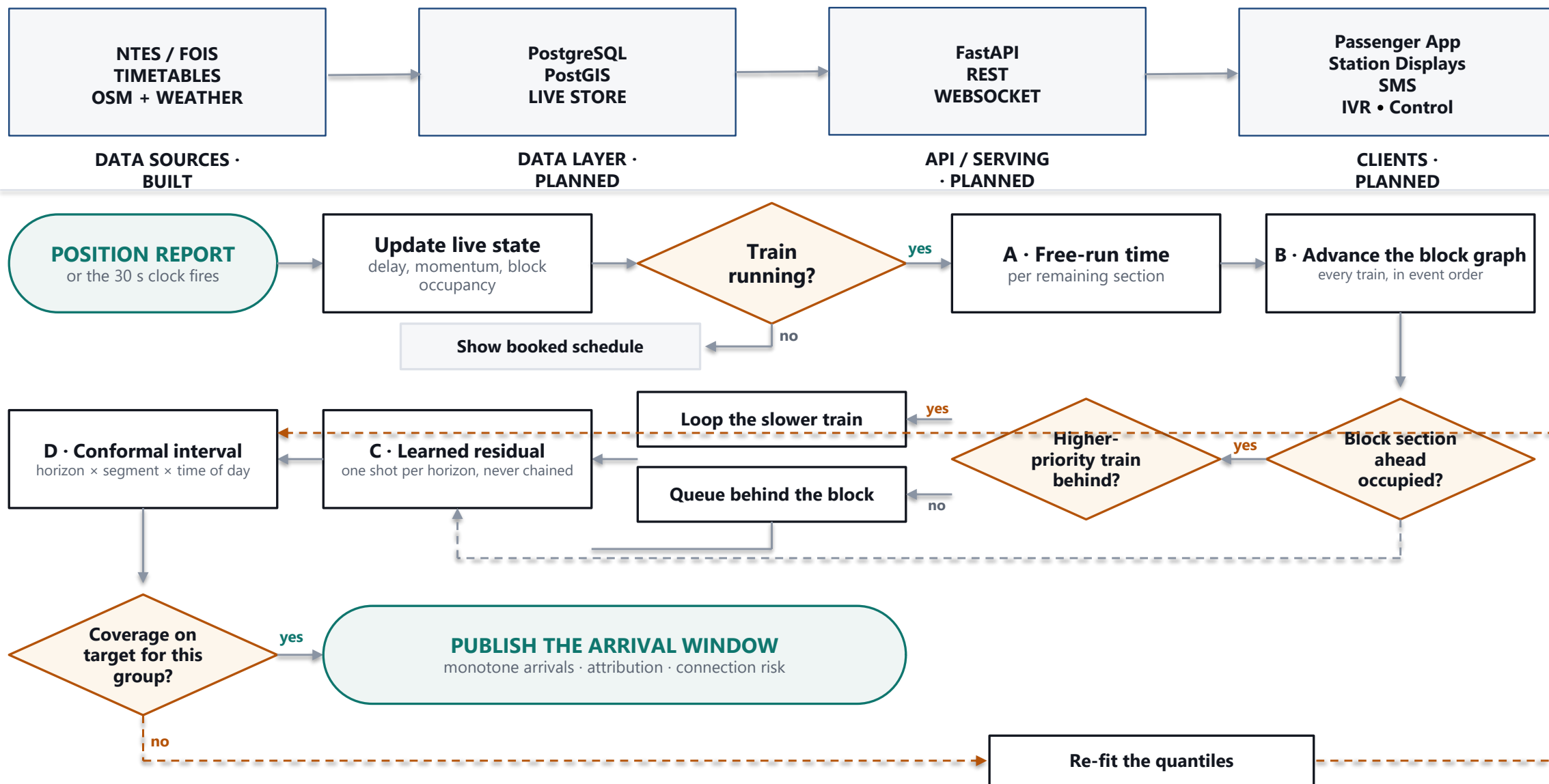


The problem — conflict and cascade



The answer — one journey re-forecast

TECHNICAL APPROACH



TECHNICAL FEASIBILITY

- Every component is mature**
LightGBM and conformal prediction are established open source.
- No new trackside hardware**
Consumes position feeds IR already produces, via a swappable adapter.
- Cheap at national scale**
One commodity node per zone; no GPU at serving time.
- The simulator is its own fallback**
With the learned layer off, rules still beat carry-forward past 3 hours.
- Validated against live running data**
499 real arrivals collected from a live feed, compared station by station.

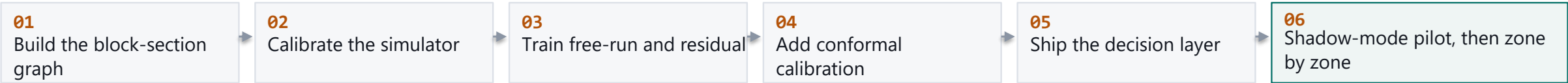
HOW WE PROVE IT

- 1 Baseline stated**
Current delay carried forward, on exactly the same rows.
- 2 Temporal holdout**
Train on earlier months, test on later. A random split leaks.
- 3 Error by horizon**
Next station, 1-3 h, 3-8 h, 8-24 h, beyond 24 h.
- 4 Coverage audited**
Against nominal 80%, by segment, horizon and time of day.
- 5 Shadow mode — planned**
Run live beside the existing ETA before a passenger sees it.

RISKS AND HOW WE RETIRE THEM

- Delay model not yet fitted**
Validated on 499 live arrivals: the tail is too heavy. Refit as data accumulates.
- Error compounds over a 40-hour run**
Direct multi-horizon residual - never chained.
- Coverage fails in the monsoon**
Mondrian calibration by segment and season, alarmed not assumed.
- Controllers distrust a black box**
Per-section attribution; advisory only, never a movement order.
- No public archive of past arrivals**
None exists, so history is accumulated by polling a live feed — already collecting.

IMPLEMENTATION ROADMAP



60%

lower mean ETA error
carry-forward 26.9 min
RAILCAST 10.9 min

62%

lower P90 error
the bad days: 77 min against 29

78.4%

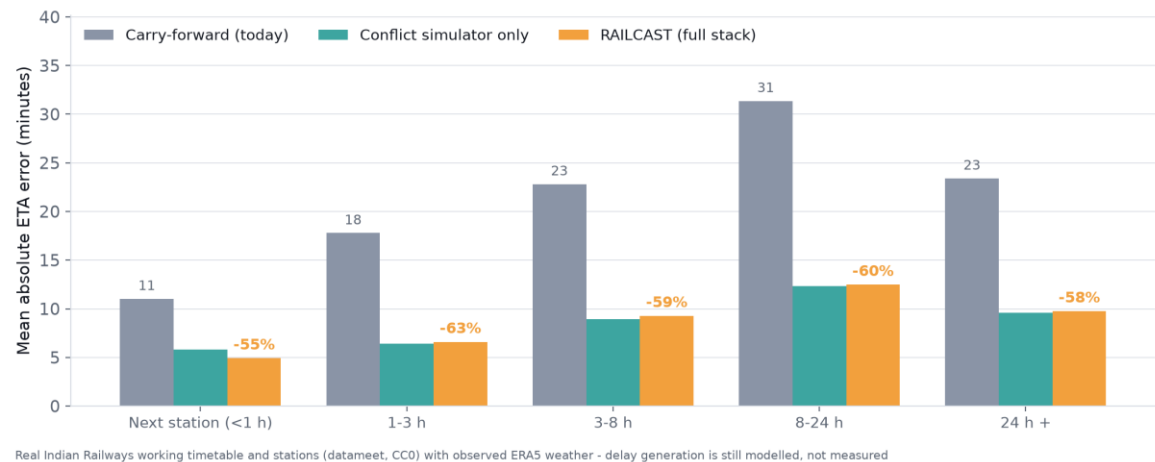
achieved coverage
of the nominal 80% window, audited per segment

0.006

reliability error
on connection-make probability

ERROR BY HORIZON, NOT ONE POOLED NUMBER

Error by forecast horizon, not one pooled number
Carry-forward is adequate for the next station. It collapses beyond a few hours - which is the horizon passengers actually plan against.



Carry-forward is adequate for the next station and collapses beyond a few hours - which is the horizon passengers actually plan against.

WHAT CHANGES AGAINST TODAY

PARAMETER	TODAY	RAILCAST
ETA method	Delay repeated forward	Per-section forecast, conflicts simulated
Horizon	The next few stops	The whole run, 40+ hours
Uncertainty	None stated	Audited 80% arrival window
Explains cause	No	Names the blocking train and section
Between stations	Blind for 40-60 min	Position inferred between stations by block-graph
Disruption	Estimate simply breaks	Re-forecast every 30 s from the current state
Decision support	None	Connection risk and costed precedence

SCALE OF DEPLOYMENT

40	Prototype corridor New Delhi - Mumbai Central, 1384 km, 202 stations
800	Zonal pilot One zone, one control office, one model shard
13,000	National Coaching trains a day, 7,300 stations, 68,000 route km

DATA SOURCES

National Train Enquiry System	live running position, via FeedAdapter
Open Government Data Platform	schedules and station master
OpenStreetMap / Overpass	block sections, junctions, loops
Open-Meteo / ERA5	historical and forecast weather, no key
IR Year Book, WTT, TSR notices	punctuality returns and restrictions

RESEARCH WE BUILD ON

- 1 R. Oneto et al., Train delay prediction systems: a big data analytics perspective, Big Data Research, vol. 11, 2018.
- 2 P. Kecman and R. M. P. Goverde, Predictive modelling of running and dwell times in railway traffic, Public Transport, vol. 7, 2015.
- 3 F. Corman and P. Kecman, Stochastic prediction of train delays in real time using Bayesian networks, Transportation Research Part C, 2018.
- 4 T. Buker and B. Seybold, Stochastic modelling of delay propagation in large networks, J. Rail Transport Planning and Management, 2012.
- 5 Y. Romano, E. Patterson and E. Candes, Conformalized quantile regression, NeurIPS 32, 2019.
- 6 V. Vovk, A. Gammerman and G. Shafer, Algorithmic Learning in a Random World, Springer, 2005 - Mondrian and conditional validity.
- 7 G. Ke et al., LightGBM: a highly efficient gradient boosting decision tree, NeurIPS 30, 2017.
- 8 The Digital Personal Data Protection Act, 2023 (India).

WHERE WE ADD TO THIS

None of the above combines a conflict-resolution simulator over the real block-section graph, a direct multi-horizon learned residual, and conditionally calibrated intervals on Indian Railways operating data.

Project links => <https://github.com/TypeError-4-ever/Railcast>